

# Do song-selective electrodes carry more song-vs-music information?

## A re-analysis of Norman–Haignere et al. (2022) with a Bellier et al. (2023) extension

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Codebase: <https://github.com/djordjeivanovic11/neuro120-project>

### Abstract

We asked a simple question: do the brain regions that respond most strongly to song also carry the most *information* a decoder could use to tell song apart from instrumental music? Norman-Haignere et al. [10] identified seven electrocorticography (ECoG) electrodes in human auditory cortex whose average response is stronger to song than to either speech or instrumental music. We re-analysed their dataset (33 electrodes, 49 stimuli, 0–2 s trial windows) and tested whether those seven electrodes actually carry more song-vs-music information than other nearby electrodes do. To make the comparison fair, we drew 7 random electrodes from the non-song pool and measured their decoder score, repeating this 1000 times. If the song electrodes are truly special, their score should beat almost all of these random groups.

They do not. A simple classifier trained on the 7 song electrodes does about as well as classifiers trained on random groups of 7 non-song electrodes ( $p = 0.36$ ), and the same holds for a geometric separation score computed from a representational dissimilarity matrix (RDM,  $p = 0.35$ ). The song electrodes only looked better than the full 33-electrode pool in the raw curves because smaller feature groups are easier for a classifier to fit, so they tend to look artificially better regardless of which electrodes you pick.

We also asked how much of the song-vs-music signal is just low-level acoustics. We subtracted out a bank of basic sound features (a cochleogram plus spectrotemporal-modulation energy) from the neural responses at every time window. Roughly two thirds of the divergence disappears, meaning most of the signal is explained by simple sound properties. About a third of the raw peak survives across all three electrode subsets (fraction surviving at peak: 0.33 song-only, 0.39 all, 0.47 no-song), hinting that the brain is tracking something beyond raw audio. At this sample size, though, the survival is only borderline significant in song-only ( $p = 0.052$ ) and does not clear significance in the other subsets ( $p = 0.108$  and  $p = 0.410$ ), so we treat it as suggestive rather than conclusive.

Finally, we tested whether the picture extends to continuous music using the Bellier et al. [1] Pink-Floyd ECoG dataset. A vocal-vs-instrumental decoder reaches balanced accuracy  $\approx 0.78$  on the full supergrid, shows the right-STG > left-STG asymmetry reported by Bellier et al. [1], and is highest on the whole electrode pool rather than any one anatomical subset. Taken together: song-selective electrodes are informative but not special, and some of the song-vs-music signal appears not to be linearly predictable from low-level acoustics.

# 1 Introduction

People separate speech, song, and instrumental music almost effortlessly. Listeners can tell voice from non-voice sounds within tens of milliseconds [8], and fMRI voxel-decomposition work has shown that music and speech sit on largely distinct cortical pathways in human auditory cortex [9]. Less is known about how the brain differentiates instrumental music from vocal music (song), which combines those two pathways.

Norman-Haignere et al. [10] took the first direct electrophysiology step toward an answer. They recorded ECoG responses to 165 natural sounds and decomposed the electrode responses into hypothesis-free components. One component responded more strongly to song than to either speech or instrumental music, and this song response could not be rewritten as a combination of the speech- and music-selective components. Seven ECoG electrodes were flagged as “song-selective.” These seven electrodes are the starting point for our project.

**What we ask.** The Norman-Haignere et al. [10] labels are defined on the *average* response magnitude: a song-selective electrode is one whose mean response is higher to song than to other sounds. But firing strongly and carrying useful information are not the same thing. Imagine a light that flashes brightly whenever there is sound of any kind: it responds strongly, but it does not help you tell song apart from music. An electrode that always fires more during song might in principle do the same thing. So we asked what happens when we probe these electrodes with a decoder instead. Specifically:

1. **Are song-selective electrodes actually better at separating song from instrumental music than other auditory-cortex electrodes?**
2. **When does the song-vs-instrumental difference appear over time, and does it stay stable?**
3. **Is the difference mostly explained by low-level acoustics, or does some of it remain after acoustics are controlled for?**

Question (1) is the central one, and it has a built-in pitfall. Smaller electrode sets are easier for a linear decoder to fit, so comparing 7 song electrodes directly against all 33 electrodes would confound subset size with selectivity. To avoid this, we built a fair comparison group. We drew 7 electrodes at random from the pool of *non*-song electrodes, measured how well those 7 decode song from music, and repeated this 1000 times. We call each such draw a *matched random subset*: same size as the song-selective group (7), but from electrodes that are *not* labelled song-selective. If the song electrodes are truly special, their decoder score should sit in the upper tail of this distribution. If not, they are doing roughly what any 7 auditory-cortex electrodes would do.

Question (3) is motivated by what is already known about the superior temporal gyrus (STG), the cortical region most electrodes in both datasets sit on. A large fraction of STG responses to complex sounds is explained by cochleogram and spectrotemporal-modulation statistics [3, 6, 8], but phonetic, prosodic, and melodic structure also contribute [7, 11, 12]. If the song-vs-music signal turns out to be a pure re-encoding of acoustics, there is nothing song-specific left to explain. If part of it survives after acoustics are controlled for, auditory cortex is tracking something beyond raw audio.

**Does it hold in continuous music?** Short, repeated stimuli are convenient for trial-averaging but may not generalise to natural listening [4]. We therefore re-run question (1) on the Bellier et al. [1] dataset: ECoG from 29 patients listening to a continuous 191-second excerpt of Pink Floyd’s *Another Brick in the Wall, Part 1*, with hundreds of electrodes per patient. This extension is deliberately bounded: one decoder, one anatomical split, one temporal-profile comparison. The goal is to check whether our central finding survives a very different stimulus regime, not to build a second project around the dataset.

## 2 Methods

### 2.1 Data

We used two publicly available human electrocorticography (ECoG) datasets. ECoG records electrical activity directly from electrodes placed on the cortical surface of neurosurgical patients, which provides high temporal resolution and good spatial coverage of auditory cortex. Both datasets provide preprocessed high-frequency brain activity (HFA), the standard ECoG proxy for local cortical firing rate.

**Norman–Haignere (2022).** Our main dataset is per-electrode HFA in 15 patients in response to 165 natural sounds, trial-averaged and sampled at 75 points across a 2-second trial window. We kept the three coarse classes relevant to our question: **song** ( $n = 11$ ), **speech** (English + foreign,  $n = 17$ ), and **music** (instrumental,  $n = 21$ ). This left 49 stimuli and 33 electrodes. Each electrode comes with a selectivity label (song, speech, or music) from Norman-Haignere et al. [10], defined on the full 165-stimulus task using trial-averaged response magnitudes.

**Bellier (2023).** Our second dataset is a 29-patient ECoG recording of people listening to a 190.72-second excerpt of a Pink Floyd song, sampled at 100 Hz. For each patient we dropped channels flagged as reference, noisy, or epileptic and attached FreeSurfer anatomical labels from the companion file. Concatenating patients along the electrode axis gives a cross-patient matrix of shape  $(T, N) = (19072, 2395)$ . We call this the *supergrid*: a single stacked matrix with 19072 time samples across 2395 electrodes pooled across all 29 patients, which is the object every downstream Bellier analysis runs on. We annotated the stimulus with a binary **vocal\_present** mask from 7 manually labelled vocal intervals covering about 58 of the 191 seconds; the song contains no other vocal content, so the non-vocal intervals are genuinely instrumental.

**Acoustic features.** To later ask how much of the neural signal is just low-level audio, we loaded two feature matrices released with Norman–Haignere: a cochleogram  $F_{\text{coch}} \in \mathbb{R}^{165 \times 6}$  (spectral energy in six frequency bands) and a spectrotemporal-modulation tensor  $F_{\text{mod}} \in \mathbb{R}^{165 \times 7 \times 9}$  (how energy moves across frequency and time). We flattened  $F_{\text{mod}}$  to  $165 \times 63$  and orthogonalised it column-wise against  $F_{\text{coch}}$  by ordinary least squares (OLS) regression, following the reference MATLAB code of Norman-Haignere et al. [10]; this prevents the two feature blocks from double-counting overlapping variance. After aligning to our 49-stimulus subset, the acoustic regressor matrix is  $A = [F_{\text{coch}} \mid F_{\text{mod, resid}}] \in \mathbb{R}^{49 \times 69}$ .

## 2.2 Analytical approach

We addressed question (1) with a linear song-vs-music decoder and a classifier-free RDM divergence score, each evaluated under the same matched 7-vs-7 random-subset control. Question (2) was addressed with time-resolved decoding and divergence, cross-temporal train-test matrices, and (for Bellier) a time-blocked binary decoder on sliding windows. Question (3) used ridge acoustic partialling at every time window, recomputing divergence on the residual. Complementary analyses were leave-one-electrode-out importance, non-linear decoder controls on Norman, and shape-matched HFA profiles across Norman and Bellier.

## 2.3 Decoding

Our first measure of song-vs-music information is a supervised decoder: given a neural feature vector, how well can a simple classifier predict the stimulus class? A high score means the information is there in a form a downstream brain region could use.

We reduce each trial to a feature vector by averaging the HFA inside a short time window, producing one value per electrode. For the “early-window” analyses we use a single  $[0.2, 0.6]$  s window. This bracket starts roughly one HFA sample after stimulus onset (past the onset transient) and ends around the peak of the classic STG category responses [3, 10], so it covers the post-onset epoch where category information is concentrated. For time-resolved analyses we use sliding windows of width 0.15 s and step 0.05 s. A 0.15 s window spans about 11 of the 75 time samples in the trial (a non-trivial local average but still short compared with the trial), and the 0.05 s step gives  $\approx 40$  overlapping windows, dense enough to draw a smooth time course.

All decoding uses  $L_2$ -penalised logistic regression with per-fold standardisation ( $C = 1$ ), the standard choice for cognitive ECoG [5]. We report *balanced accuracy* (bacc), the mean per-class recall.<sup>1</sup> Cross-validation (CV) uses stratified group 5-fold splits with stimulus ID as the grouping variable, repeated 20 times with different random seeds for the headline numbers. Grouping by stimulus is essential: without it, the classifier could cheat by memorising a stimulus it saw during training, inflating accuracy without reflecting class-level information.

## 2.4 Divergence: a classifier-free separation score

Decoding asks whether a classifier *can* separate the classes. We also wanted a geometric measure, independent of any classifier, that asks how structured the neural responses are in the first place. For this we use a representational dissimilarity matrix (RDM): the matrix of pairwise distances between the neural responses to every pair of stimuli. If category is reflected in neural geometry, pairs of stimuli from different classes should sit farther apart in the RDM than pairs from the same class.

Let  $M \in \mathbb{R}^{N_s \times N_e}$  be the window-averaged feature matrix (one row per stimulus, one column per electrode). We  $z$ -score each column so no single electrode dominates the distance, and define the RDM  $R$  by correlation distance between stimulus rows,  $R_{ij} = 1 - \text{corr}(z(M_i), z(M_j))$ . Correlation distance compares response *patterns* rather than absolute magnitudes, which is a standard choice in representational similarity analysis (RSA). For a pair of classes  $(a, b)$  the *divergence* is the mean

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<sup>1</sup>We use balanced rather than plain accuracy because the classes are unequal in size; a lazy classifier that always guesses **music** would look deceptively good on plain accuracy.

between-class pairwise distance minus the average within-class distance:

$$\mathcal{D}(a, b) = \underbrace{\overline{R_{ab}}}_{\text{between}} - \frac{1}{2} \left( \underbrace{\overline{R_{aa}}}_{\text{within-}a} + \underbrace{\overline{R_{bb}}}_{\text{within-}b} \right), \quad (1)$$

where  $\overline{R_{ab}}$  is the mean of the off-diagonal block between labels  $a$  and  $b$ , and  $\overline{R_{aa}}$  is the mean of the strictly upper-triangular entries inside the class- $a$  block. In plain words: we compare *how different are song responses from music responses* against *how different are songs from other songs, and music from other music*. A positive divergence means cross-class stimuli are farther apart than same-class stimuli, so the classes are genuinely separated in neural space. We use this score rather than, say, decoder margin because it does not need a classifier, and because its numerator and denominator live on the same distance scale, so changes in the number map cleanly onto geometric changes in the RDM.

## 2.5 Significance testing

Every headline number is backed by one or more of three controls. Each asks a different question about where the effect could have come from by chance.

1. **Bootstrap over stimuli.** We resample stimulus IDs within-class with replacement, deduplicate to avoid zero-distance self-pairs, recompute the divergence curve, and report a percentile 95% confidence interval (CI) across  $B = 1000$  resamples. This tells us how reliable the number is across resamples of the stimuli.
2. **Label-shuffle permutation.** To build a chance baseline, we randomly reassign the class labels among the stimuli (within the  $\{a, b\}$  pair only) and recompute the divergence 1000 times. At each time window we record the 95th percentile of these shuffled divergences, which we call the *chance-level band*. Any curve sitting below this band is indistinguishable from what you would get with random labels. The peak  $p$ -value is  $(\#\{\text{null peak} \geq \text{obs peak}\} + 1) / (n_{\text{perm}} + 1)$ .
3. **Matched random-subset control.** The song-selective set has only 7 electrodes, so we cannot compare it directly against all 33 electrodes without confounding group size with selectivity. Instead, we build a fair baseline by drawing 7 electrodes at random (without replacement) from the non-song pool, recomputing bacc and divergence, and repeating 1000 times. We then compare the observed song-only value to this distribution. If song-selective electrodes carry privileged information, the song-only score should sit in the upper tail; if not, it sits somewhere in the middle.

## 2.6 Controlling for low-level acoustics

Songs and instrumental music sound different at the most basic level: songs have voices, different pitch patterns, different rhythms. So any brain region that separates song from music might just be picking up on those basic sound differences, not anything deeply “musical.” To test this we use a technique called linear partialling: we predict the neural signal from a bank of basic sound features, subtract out that prediction, and ask whether song-vs-music geometry remains in the residual. Anything left is, by construction, not explainable by a linear readout of those features.

Concretely, at every sliding window we fit a ridge regression from the acoustic features  $A$  to the neural matrix  $M$ , then replace  $M$  by the residual: what the neural signal looks like *after* subtracting

out the predicted acoustic contribution. With centred columns  $\tilde{A} = A - \bar{A}$  and  $\tilde{M}_0 = M - \bar{M}$ , the ridge coefficients  $\hat{\beta}$  and the residual  $\tilde{M}$  are

$$\hat{\beta} = \left( \tilde{A}^\top \tilde{A} + \lambda I \right)^{-1} \tilde{A}^\top \tilde{M}_0, \quad \tilde{M} = \tilde{M}_0 - \tilde{A} \hat{\beta}. \quad (2)$$

The ridge penalty  $\lambda$  is there for numerical stability:  $A$  has 69 columns and only 49 rows, which makes the uncorrected inverse ill-conditioned. Very small  $\lambda$  lets the regression chase noise; very large  $\lambda$  under-removes real acoustic structure and biases the partialled divergence upwards. We use  $\lambda = 1$  as a moderate compromise and confirmed on a small sweep that the qualitative pattern (surviving fraction  $\approx 1/3$ , similar ordering across subsets) is stable for  $\lambda \in [0.1, 10]$ . The RDM and divergence are then recomputed from  $\tilde{M}$ . For the permutation test we re-fit the regression inside every shuffle rather than fitting once. This ensures the null reflects neural geometry *after* acoustics are controlled for, rather than inheriting a shortcut whenever the acoustic features themselves carry class information.

## 2.7 Cross-temporal generalisation

So far we have asked when during the trial the song-vs-music distinction is strongest. A deeper question is whether the brain uses the *same* neural code the whole time, or whether the pattern shifts as the sound plays. A stable code throughout the trial is the signature of a sustained categorical representation; a drifting code is the signature of an onset-driven one. We test this by training a classifier at one time and testing it at another: if a classifier trained at  $t_i$  still works at  $t_j$ , the code is stable across that interval; if it fails, the code at  $t_i$  is a different representation from the one at  $t_j$ . We train  $L_2$  logistic regression on the song-vs-music task at each source window  $t_i$ , then test it at every destination window  $t_j$  on held-out stimuli from the same CV split. Stacking all  $(t_i, t_j)$  pairs gives an  $N_t \times N_t$  bacc matrix: bright off-diagonals mean a stable code, dark off-diagonals mean it drifts. For a compact summary we compare the mean of the diagonal ( $t_i = t_j$ ) against the mean of off-diagonal entries with  $|t_i - t_j| \geq 0.25$  s. We use 0.25 s as a conservative buffer (several sliding-window widths) so the off-diagonal mean does not inherit same-window overlap.

## 2.8 Leave-one-out

The random-subset control tests song electrodes as a group. A complementary question is whether any individual electrode contributes disproportionately, and whether those contributions concentrate in the song group. For each electrode  $i$  we compute how much performance drops when it is removed:  $\Delta_i = \text{bacc}(\text{all}) - \text{bacc}(\text{all} \setminus \{i\})$ , and  $\Delta_i^{\mathcal{D}}$  the same way for divergence. To test whether song electrodes contribute more than others on average, we run a group-label permutation test with 1000 shuffles, comparing the mean  $\Delta$  inside the song group against the mean outside. We prefer a permutation test over a  $t$ -test here because the per-electrode  $\Delta$  values are small, non-normal, and have unequal variance across groups.

## 2.9 Bellier extension: vocal-vs-instrumental decoding

To check whether the central finding generalises beyond short repeated clips, we re-ran a simpler version of our decoder on Bellier. Instead of three coarse classes, we tested the binary question the stimulus allows: can we decode when there is vocal content versus when there is not? This also lets us test the anatomical claim of Bellier et al. [1] that the vocal signal is stronger in right STG than left STG.

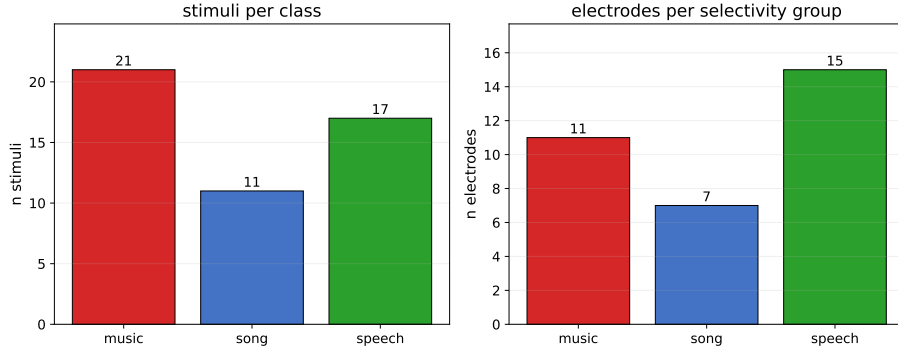
We tiled the supergrid with 500 ms windows at 100 ms step (1903 windows in total) and labelled each window `vocal` if more than 50% of its samples fell inside a manually annotated vocal interval. A 500 ms window is long enough to span a sung syllable or two (the natural unit of the vocal/instrumental contrast in continuous song) while still being short compared with the 191-second song. The 100 ms step gives dense temporal coverage without creating near-duplicate windows. The 50% threshold is a simple majority rule: a window straddling a vocal/instrumental boundary is assigned to whichever class dominates it. We analysed four electrode subsets: `all` (the full 2395-electrode supergrid), `right_STG`, `left_STG`, and `non_STG`. The STG subsets use FreeSurfer `superiortemporal` labels split by the `ctx-lh` / `ctx-rh` prefix.

**Blocked-time CV with interleaved folds.** Continuous-stimulus ECoG needs a time-blocked CV scheme because adjacent samples are correlated, so random splits would leak information between train and test. A contiguous 5-fold split is the obvious first try, but it breaks here: the vocal annotations cover only the first  $\approx 92$  s of the 191 s song, so folds 3 and 4 of a contiguous split fall entirely in the instrumental region and balanced accuracy is undefined on them. We fix this with *interleaved blocked folds*. The intuition is like dealing cards: we chop the song into 15 shorter contiguous blocks and deal them into 5 folds, so each fold is the union of three non-adjacent segments that are spread across the recording. Formally, we divide time into  $n_{\text{folds}} \times n_{\text{blocks}/\text{fold}} = 15$  equal contiguous blocks and assign block  $b$  to fold  $b \bmod n_{\text{folds}}$ . Every fold then contains both vocal and instrumental windows, no single 500 ms window crosses a fold boundary, and within-segment autocorrelation is preserved.

On these folds we fit  $L_2$  logistic regression with `LogisticRegressionCV` ( $C \in \{10^{-3}, \dots, 10^3\}$ , inner 3-fold) on the electrode-means of each window. We report both balanced accuracy and the area under the ROC curve (ROC-AUC), the latter being a threshold-free summary of how well the decoder’s continuous scores separate the two classes. We only run a more expensive non-linear model on subsets that already show a clear linear signal, so our small CNN (`TinyTemporalCNN`: two 1-D convolutions, adaptive pooling, linear head, 40 training epochs) is restricted to subsets whose logistic regression clears chance by at least 0.05 bacc. This avoids tuning a complex model against near-chance features, where apparent gains are most likely to reflect noise. The 40-epoch schedule is a standard short-training budget for a small model on the few thousand windows we have; we did not observe further improvement beyond this.

## 2.10 Cross-dataset temporal profiles

The Bellier decoder tells us the vocal-instrumental distinction is there, but not how the response looks over time. To connect the two datasets qualitatively, we compared average HFA profiles around each event of interest. For Bellier, we computed event-locked averages in  $[-0.5, 1.5]$  s around every vocal and instrumental onset, averaged across electrodes in each anatomical subset. For Norman–Haignere we used trial-locked averages over the 2-second trial window, averaged across selectivity group and stimulus class. Because the two datasets are locked to different kinds of events, we compared them only on response shape, not amplitude. For each profile we extracted peak latency, an onset-sustained ratio (mean in  $[0, 0.2]$  over mean in  $[0.5, 1.5]$ ; a ratio below 1 means the response is sustained), and a trapezoidal area under the curve over  $[0, 1.5]$  s (AUC here means “integrated response magnitude,” not the ROC-AUC used by the decoder). The  $[0, 0.2]$  s bin covers the short post-event transient reported in the STG onset/sustained map of Hamilton et al. [3], and  $[0.5, 1.5]$  s covers the later sustained phase



**Figure 1: Dataset layout.** Stimulus counts per coarse class and electrode counts per selectivity group after trimming to the 49-stimulus {song, speech, music} subset. The three subsets used later are *all* (33 electrodes), *song\_only* (7 song-selective electrodes), and *no\_song* (26 speech- and music-selective electrodes); *matched random subsets* are 1000 draws of 7 electrodes from the *no\_song* pool for a fair size-matched null.

once the transient has decayed; the ratio of the two is the single number that most cleanly separates transient onset electrodes from sustained ones.

### 2.11 Non-linear controls

Finally, we wanted to rule out the possibility that our negative random-subset result is an artefact of using a linear decoder. If a non-linear model could extract much more song-vs-music information than logistic regression, the song electrodes might still dominate under that model. We ran two non-linear alternatives on the same grouped CV splits. The first is a kernel SVM with a radial basis function (RBF) kernel, a standard non-linear baseline that captures smooth non-linear decision boundaries. The second is a frozen autoencoder bottleneck (a 3-layer multilayer perceptron, or MLP, with a 16-unit hidden code, trained for 40 epochs with mean squared error on training-fold features only) followed by logistic regression on the 16-unit code; we froze the autoencoder before training the logistic stage so the two stages are not optimised jointly. The 16-unit bottleneck and 40-epoch schedule are small relative to our feature dimension (33 electrodes) and dataset (49 stimuli), chosen so the autoencoder compresses rather than memorises.

## 3 Results

### 3.1 Decoding song, speech, and music

*Can a simple classifier tell song, speech, and music apart at all using these electrodes?* If it cannot, there is no point asking subtler questions. Figure 1 summarises the dataset layout after class trimming: 33 electrodes, 49 stimuli, and 75 time samples spanning 0–2 s, with (11, 17, 21) stimuli per class and (7, 15, 11) electrodes per selectivity group. A three-class logistic-regression decoder separates them easily: balanced accuracy reaches 0.97 on the early [0.2, 0.6] s window (macro-F1 0.97; per-class recall 1.00, 0.91, 1.00 for music, song, speech). Crucially, every residual error is a song-to-music confusion (panel (a) of Fig. 2). That specific failure mode is what the rest of the paper is about.



**Table 1: RDM divergence summary.** Peak divergence, peak time, early-window mean ([0.2, 0.6] s), and permutation  $p$ -value at peak (1000 shuffles).

| subset    | peak $\mathcal{D}$ | peak time (s) | early mean $\mathcal{D}$ | $p_{\text{peak}}$ |
|-----------|--------------------|---------------|--------------------------|-------------------|
| all       | 0.215              | 1.72          | 0.167                    | 0.001             |
| no_song   | 0.108              | 0.27          | 0.049                    | 0.005             |
| song_only | 0.408              | 0.37          | 0.299                    | 0.002             |

### 3.2 When does the song-vs-music difference appear?

*We now zoom in on the song-vs-music pair specifically: when during the trial is that distinction actually decodable?* The answer: throughout the trial, at essentially every electrode subset (panel (b) of Fig. 2). Peak balanced accuracy is 1.00 on **all** and on the 7 song-selective electrodes (**song\_only**), and 0.95 on the remaining 26 speech- and music-selective electrodes (**no\_song**). Timing differs across subsets: the full pool (**all**) peaks at 0.32 s, while **no\_song** and **song\_only** both peak later at 0.67 s. The cleanest thing to say about the full-pool early peak is that it relies on combining information across all 33 electrodes; neither subset on its own reaches a decoder peak that early. These timings are consistent with a contribution from early-onset electrodes in the STG onset-vs-sustained spatial map of Hamilton et al. [3]. The decoder clearly does not depend on song-selective electrodes to reach chance-clearing performance: **no\_song** still reaches 0.95. Whether **song\_only** outperforms **no\_song** in a meaningful way is exactly the question the random-subset control in §3.4 is designed to answer.

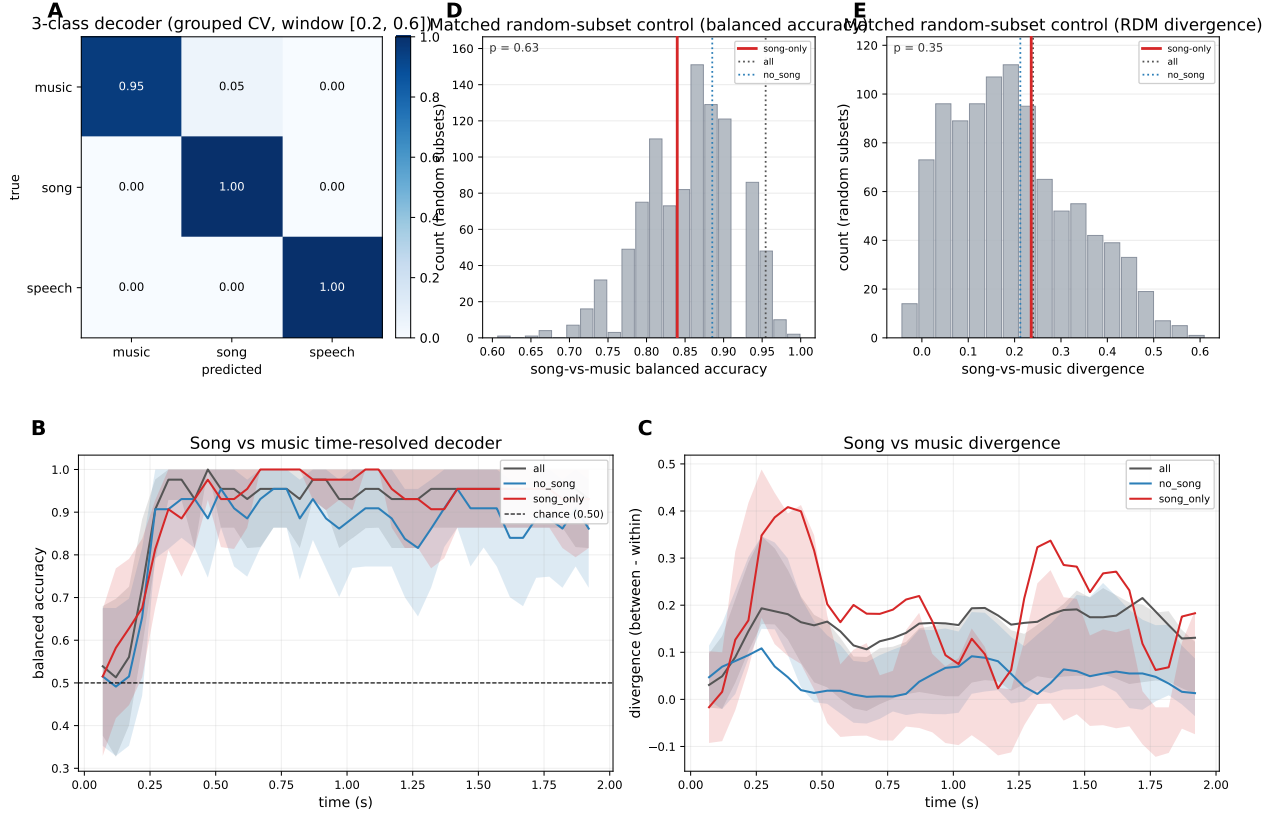
### 3.3 Neural separation of song and music

*Decoding accuracy tells us a classifier can separate the classes. But is the separation also visible in the raw neural geometry, with no classifier involved?* We cross-checked the decoder result with the classifier-free divergence score from §2.4. Every subset produces a divergence curve that rises above the chance-level band (what the score looks like under shuffled labels) within the first 0.3 s (panel (c) of Fig. 2; Table 1). Peak divergence is largest for **song\_only** (0.408,  $p_{\text{peak}} = 0.002$ ), though it is also the noisiest curve, as expected from only 7 electrodes. **no\_song** and **all** peak lower in absolute terms but still clear permutation  $p < 0.01$ . Song-vs-music structure is therefore present in the neural geometry of every subset, not only in the decoder. As in panel (b) of Fig. 2, the **song\_only** curve looks larger than **all** and **no\_song**, but the apparent **song\_only** advantage here is again confounded with subset size; the fair comparison is in §3.4.

### 3.4 Random-subset control: are song-selective electrodes special?

*This is the central question of the paper.* The raw curves in panels (b–c) of Fig. 2 look as if **song\_only** beats the larger **no\_song** pool, which would suggest that song-selective electrodes are uniquely informative. But comparing against all 33 electrodes is not a fair test, because smaller electrode sets are easier to decode from. The fair comparison is against other groups of 7 electrodes drawn at random from the non-song pool.

Song-only is not special under this fair comparison (panels (d–e) of Fig. 2). On balanced accuracy it sits at the 64th percentile of the 1000 random draws (0.885,  $p = 0.36$ ), and at a similar percentile on divergence (0.236,  $p = 0.35$ ).



**Figure 2: Norman-Haignere decoding, geometry, and matched random-subset null.** (a) Row-normalised confusion matrix for the 3-class logistic-regression decoder (grouped 5-fold CV, early window [0.2, 0.6] s). (b) Time-resolved song-vs-music balanced accuracy (0.15 s sliding windows; shaded: bootstrap 95% CI; dashed: chance). (c) Classifier-free song-vs-music RDM divergence (same windows; shaded: bootstrap CI). (d-e) Matched random-subset null for balanced accuracy and divergence: grey histograms are 1000 draws of 7 non-song electrodes; red: observed `song_only`; reference lines for `all` and `no_song` pools. Empirical one-sided  $p$ -values vs. the random draws are printed on the histograms.

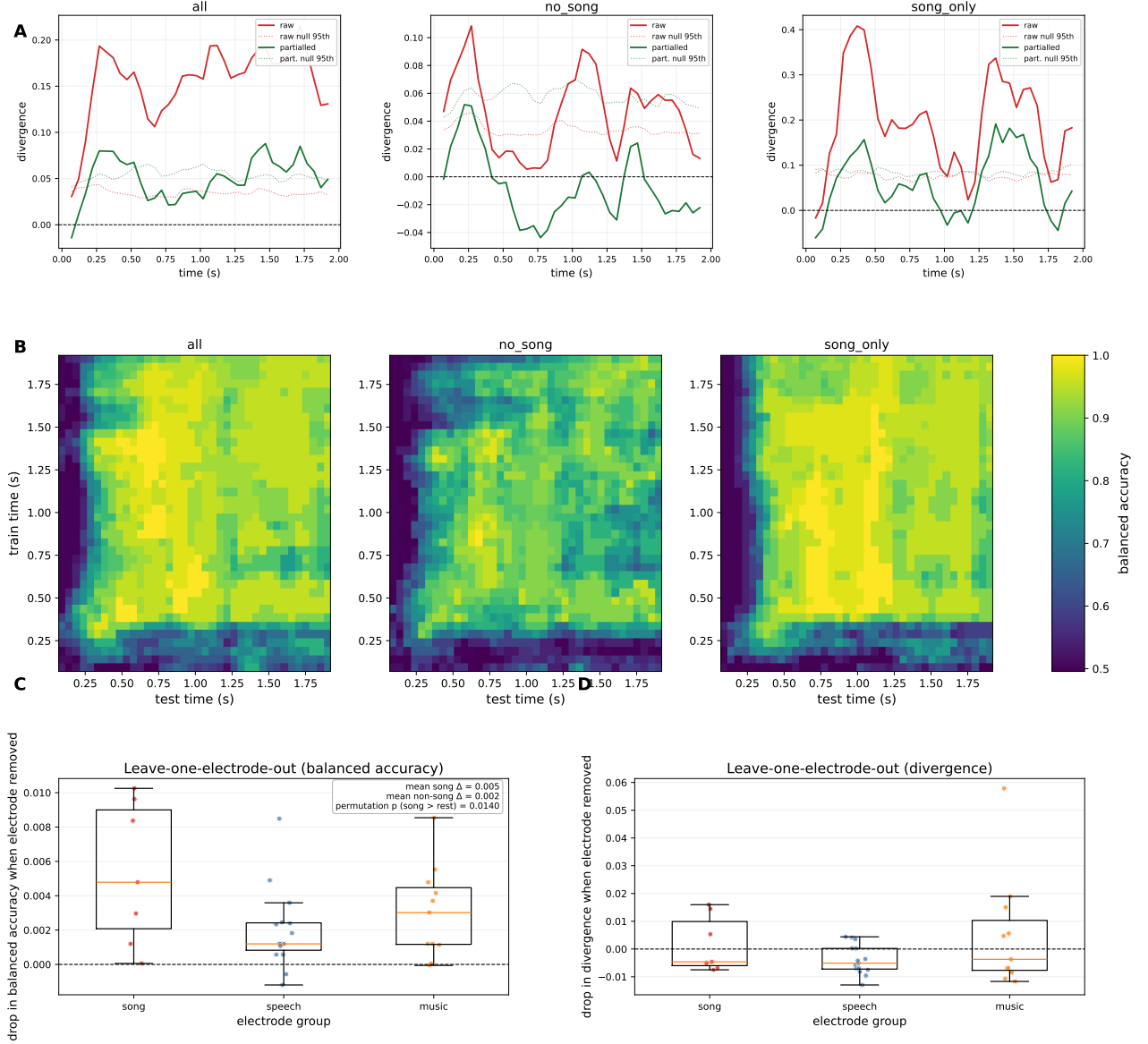
Two numbers in this result might seem to disagree at first, so it is worth walking through what each one means. We look at both the *tail* of the random-draw distribution (via the  $p$ -value) and the *average* of that distribution (via a bootstrap 95% CI on the song-only minus random-draw mean). The  $p$ -values (0.36, 0.35) answer: *does the song-only score lie in the top few percent of random-group scores?* The answer is no; it sits near the middle. The bootstrap CI for song-only minus random-draw mean is strictly positive (about +0.03 on bacc, +0.04 on divergence), which answers a different question: *is the song-only score above the average of the random draws?* The answer is yes, very slightly. Both can be true: there is a small per-electrode advantage on song-selective electrodes, but it is well within what any 7 randomly chosen non-song electrodes can produce. This is our central negative result: song electrodes as a group do not carry disproportionately more song-vs-music information than neighbouring auditory-cortex electrodes.

**Why does song-only appear to beat all-electrodes?** The reason is a size effect, not a selectivity effect. A logistic regression with  $k = 7$  features is better conditioned than one with  $k = 33$ : fewer features, gentler regularisation, lower cross-fold variance. Any random 7-electrode non-song subset gets the same benefit, which is exactly what panels (d–e) of Fig. 2 show. The correct population-level benchmark is therefore the full 33-electrode pool, and the song-only advantage over it is a small-subset ceiling effect.

**Are we just measuring the labels?** One might worry that because the song electrodes were defined on the same dataset, our result is circular. This concern is partly mitigated, not fully eliminated. Norman-Haignere et al. [10] derived their labels from the full 165-stimulus task using trial-averaged responses, while our decoder and divergence score are computed on a different object: per-electrode responses to the 49-stimulus {song, speech, music} subset. So the statistical target is not identical. But the labels are still inherited from the same dataset and same electrodes, so we cannot rule out circularity on principle. The best argument against it here is empirical: if the labels really drove our result, the song electrodes should have sat in the extreme tail of the random-draw distribution. They did not.

### 3.5 What remains after controlling for acoustics?

*If song electrodes are not disproportionately informative, the natural follow-up is: what is the decoder picking up on in the first place – just basic acoustics, or something beyond them?* The acoustic-partialled analysis (§2.6) answers this by recomputing divergence after subtracting out what is linearly predictable from the cochleogram and spectrotemporal-modulation features. A non-trivial fraction of the song-vs-music signal survives the acoustic subtraction: 33–47% of the raw peak divergence across the three subsets (panel (a) of Fig. 3; Table 2). Statistically, this surviving structure is borderline. The partialled **song-only** peak sits right at the conventional threshold ( $p = 0.052$ ); the other two subsets do not clear it ( $p = 0.108$  for **all**,  $p = 0.410$  for **no-song**). What we can safely say is narrower than “post-acoustic code”: some structure remains after regressing out this particular linear acoustic feature set, and the amount that remains is similar across the three subsets. That pattern is in the direction expected for integrative coding in STG [7, 11, 12] – i.e., the STG combining acoustic input with higher-order category cues – but at this sample size and with this feature bank we cannot claim a robustly significant non-acoustic component.



**Figure 3: Mechanistic follow-ups on Norman-Haignere.** (a) Acoustic-partialled divergence: red, raw divergence with label-shuffle 95th-percentile band; green, divergence after ridge-regressing cochleogram + spectrotemporal-modulation features out of the neural features at each window (partialled band refits the regression inside every shuffle). Columns: *all*, *no\_song*, *song\_only*. (b) Cross-temporal generalisation (song vs music): balanced accuracy when training at  $t_i$  and testing at  $t_j$ , shared colour scale across subsets. (c-d) Leave-one-electrode-out change in time-mean balanced accuracy and in early-window divergence; box + jitter by selectivity group; annotation gives group means and permutation  $p$  for song > rest on  $\Delta_{\text{bacc}}$ .

**Table 2: Acoustic partition.** Raw vs acoustic-partialled divergence, fraction of the raw peak that survives partialling, and permutation  $p$ -values (1000 shuffles, ridge  $\lambda = 1$ , regressors  $A_{\text{full}} \in \mathbb{R}^{49 \times 69}$ ).

| subset    | peak raw | peak partial | surviving fraction | $p_{\text{partial}}$ |
|-----------|----------|--------------|--------------------|----------------------|
| all       | 0.215    | 0.088        | 0.395              | 0.108                |
| no_song   | 0.108    | 0.052        | 0.470              | 0.410                |
| song_only | 0.408    | 0.191        | 0.331              | 0.052                |

**Table 3: LOO group summary.** Mean drop in time-mean bacc and in early-window  $\mathcal{D}$  when each electrode is removed; permutation  $p$  for song-vs-rest on  $\Delta\text{bacc}$  (1000 shuffles).

| group  | $n$ | mean $\Delta\text{bacc}$ | mean $\Delta\mathcal{D}$ | $p$ (song vs rest, bacc) |
|--------|-----|--------------------------|--------------------------|--------------------------|
| song   | 7   | +0.0053                  | +0.0017                  | 0.014                    |
| music  | 11  | +0.0030                  | +0.0049                  | —                        |
| speech | 15  | +0.0020                  | −0.0039                  | —                        |

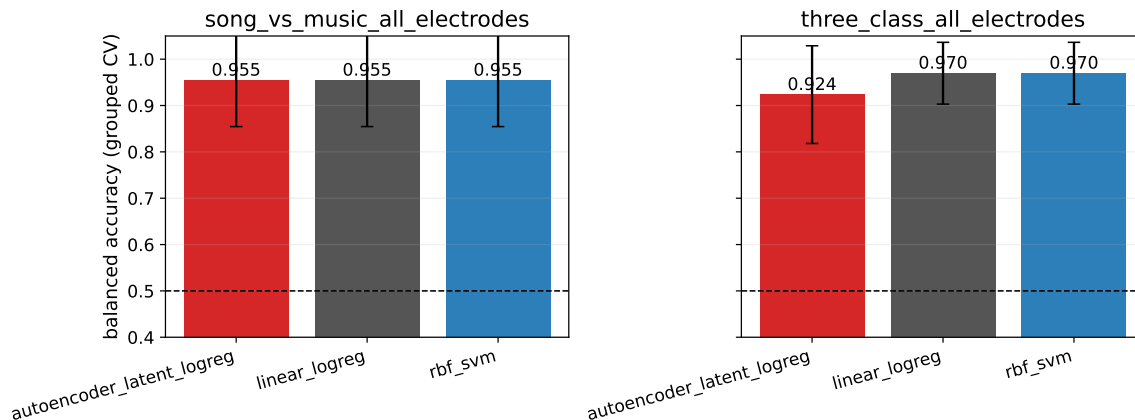
### 3.6 Temporal stability of the song-vs-music code

*Even if song electrodes are not more informative at peak, they might carry a qualitatively different **kind** of code: one that stays stable across the trial rather than drifting.* Cross-temporal generalisation tests this: a decoder trained at time  $t_i$  is tested at every other time  $t_j$ , and bright off-diagonals indicate a stable code. A sustained categorical code should generalise across time; a purely onset-driven code should not [3, 5].

The song-selective subset shows a somewhat more temporally stable representation than the non-song pool (panel (b) of Fig. 3). Diagonal vs. off-diagonal-at- $\geq 0.25$  s balanced accuracy is 0.92/0.84 for all, 0.92/0.83 for **song\_only**, and 0.87/0.76 for **no\_song**: a decoder trained at 0.3 s still works at 1.5 s on the song electrodes, and less so on the non-song pool. We do not have a permutation null on this comparison, so we read the  $\approx 8$ -point vs.  $\approx 11$ -point off-diagonal gaps as suggestive of a sustained categorical code on the song electrodes rather than a formal significant result.

### 3.7 Per-electrode contributions (leave-one-out)

*The random-subset control in §3.4 tests song electrodes as a group. Here we run a complementary test at the single-electrode level, asking whether removing any one electrode disproportionately hurts the decoder, and whether those important electrodes concentrate in the song group.* Song-labeled electrodes show a slightly higher mean drop in time-averaged balanced accuracy when removed than the other groups (panels (c–d) of Fig. 3; Table 3). Mean  $\Delta\text{bacc}$  is +0.0053 for song, +0.0030 for music, and +0.0020 for speech, with a song-vs-rest permutation  $p = 0.014$  (1000 shuffles). The magnitudes are small and the distributions overlap, so we read this as a modest single-electrode tilt, not a second “special population” claim: the fair 7-vs-7 matched subset test in §3.4 remains the primary evidence that the seven song contacts as a *set* are not unusually informative relative to other size-matched non-song draws.



**Figure 4: Non-linear supplement.** Grouped CV balanced accuracy of three model families on two tasks (three-class and song-vs-music, all-electrodes) using identical stimulus-grouped 5-fold splits. Error bars are across-fold standard deviations. Neither the RBF-SVM nor the autoencoder-latent pipeline materially outperforms the linear logistic baseline.

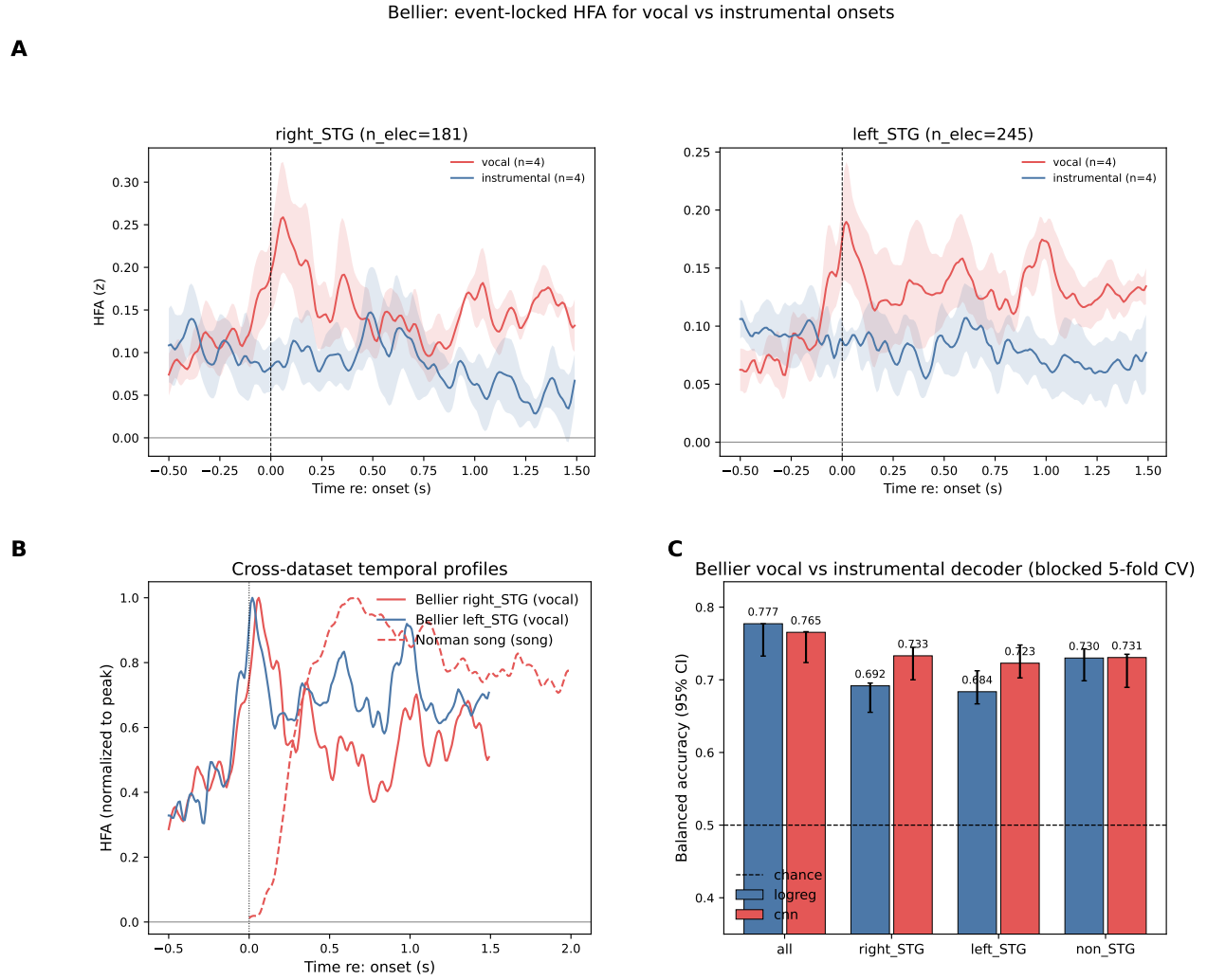
### 3.8 Do more complex models help?

A concern with our negative random-subset result is that it might be an artefact of using a linear decoder: maybe a non-linear model could extract much more song-vs-music information from the song electrodes. Non-linear models do not beat the linear baseline on this dataset (Fig. 4). Neither an RBF-SVM nor an autoencoder-latent + logreg pipeline improves on logistic regression:  $\Delta\text{bacc} = 0$  for song-vs-music and  $\Delta\text{bacc} \in \{-0.046, 0\}$  for the three-class task. At this sample size and feature dimension, a simple linear decoder already extracts most of the available signal, so adopting a linear decoder throughout the paper does not seem to be leaving obvious non-linear information on the table. This is consistent with the STG literature on low-dimensional feature tuning [6, 8].

### 3.9 Does the result hold in continuous music?

So far we have shown that song-vs-music structure is decodable and visible in geometry for every Norman subset, but the seven song-selective contacts are not privileged under a fair 7-vs-7 draw; that much of the signal overlaps our acoustic feature bank, with a suggestive partialled residual; and that non-linear decoders do not overturn the linear picture on Norman. We next ask whether the same qualitative story—strong decodability without a privileged anatomical subset—holds in continuous, naturalistic music using Bellier et al. [1]. Table 4 reports the vocal-vs-instrumental decoder on the 2395-electrode cross-patient supergrid defined in §2.9. The interleaved-folds fix was essential: with naive contiguous folds, folds 3 and 4 of a 5-fold split fall entirely in the instrumental region and balanced accuracy is undefined.

Two findings stand out in Table 4 and panel (c) of Fig. 5. First, right STG slightly beats left STG on both logreg and CNN, going in the same direction as the right-hemispheric vocal bias reported by Bellier et al. [1]. The absolute difference is small (0.692 vs. 0.684 on logreg), and we did not run a permutation null for this subset comparison, so we treat this as a directional replication rather than a formally significant finding. Second, the whole-pool bar is the highest overall, consistent with distributed rather than STG-exclusive coding. The CNN gains about 4 bacc points on the STG subsets



**Figure 5: Bellier extension and cross-dataset profiles.** (a) Event-locked mean HFA (SEM band) at vocal and instrumental onsets for right and left STG ( $n = 4$  onset events per condition after the event detector). (b) Peak-normalised overlay: Bellier STG at vocal onsets vs. Norman song-electrode responses on song trials (shape comparison only; different normalisations and locking). (c) Vocal-vs-instrumental decoder: mean balanced accuracy with 95% bootstrap CI for logistic regression and TinyTemporalCNN (Table 4).

**Table 4: Bellier (2023) vocal-vs-instrumental decoder.** 1903 windows, 500 ms / 100 ms, interleaved 5-fold blocked-time CV. CNN is run only on subsets where logreg clears chance by  $\geq 0.05$ .

| subset    | $n_{\text{elec}}$ | logreg bacc  | logreg AUC | CNN bacc     | CNN AUC |
|-----------|-------------------|--------------|------------|--------------|---------|
| all       | 2395              | <b>0.777</b> | 0.898      | 0.765        | 0.839   |
| right_STG | 181               | 0.692        | 0.814      | <b>0.733</b> | 0.839   |
| left_STG  | 245               | 0.684        | 0.780      | 0.723        | 0.806   |
| non_STG   | 1969              | 0.730        | 0.892      | 0.731        | 0.825   |

but not on the whole-pool or non-STG subsets, suggesting the within-window temporal structure that the window-average discards is concentrated in STG.

### 3.10 Comparing response shapes across datasets

*The Bellier decoder shows the vocal-vs-instrumental distinction is decodable, but it does not tell us how the underlying response looks over time. We now compare the shape of that response against Norman to see whether the two datasets are emphasising the same or different phases of the HFA.* We compared HFA profiles on response shape only, not amplitude (panels (a–b) of Fig. 5). The cleanest summary is the onset-sustained ratio (Table 5). Bellier right\_STG at vocal onsets is transient (ratio 1.57), Bellier left\_STG is balanced (1.11), and Norman song electrodes at song trials are strongly sustained (0.11), which is the most sustained response in the table. The two datasets sample different phases of the response: Bellier highlights a short onset transient while Norman highlights a long sustained phase.

**Table 5: Temporal-profile features (selected).** Peak latency, onset-sustained ratio (mean  $[0, 0.2]$ /mean  $[0.5, 1.5]$ ;  $< 1$  = sustained), and trapezoidal AUC in  $[0, 1.5]$ . Lower rows: Norman group  $\times$  stimulus class.

| dataset | profile                  | peak $t$ (s) | onset/sustained | AUC    |
|---------|--------------------------|--------------|-----------------|--------|
| Bellier | right_STG @ vocal        | 0.06         | 1.57            | 0.23   |
| Bellier | right_STG @ instrumental | 0.49         | 1.30            | 0.12   |
| Bellier | left_STG @ vocal         | 0.02         | 1.11            | 0.20   |
| Norman  | song-elec @ song         | 0.66         | <b>0.11</b>     | 100.75 |
| Norman  | song-elec @ music        | 0.40         | 0.44            | 34.72  |
| Norman  | music-elec @ music       | 0.37         | 0.43            | 82.08  |
| Norman  | speech-elec @ speech     | 0.35         | 0.49            | 122.41 |

This shape mismatch is expected given how differently the stimuli are presented. Norman’s 2-second clip paradigm trial-locks responses, which averages in the sustained post-onset phase; Bellier’s event-locking at vocal onsets inside a continuous song samples mostly the transient at stimulus change. The sustained song-tracking shape is clearly visible in Norman song electrodes, but it is not the first-order phenomenon at Bellier’s right-STG vocal onsets. For our central claim (song electrodes are informative but not privileged), this matters because it shows the claim does not depend on a specific paradigm: both the trial-locked and the event-locked view return a decodable signal that is distributed rather than concentrated in any one anatomical subset.



## 4 Discussion

**How this refines Norman-Haignere et al. [10].** The original paper showed that song-selective electrodes exist in the sense that their trial-averaged response is larger for song than for other sounds. Our re-analysis asks a slightly different question: are those same electrodes disproportionately *informative* for a decoder trying to separate song from instrumental music? An electrode can fire more strongly during song without actually helping a classifier, so the two questions are not the same. The answer we get is: no, under the primary matched 7-vs-7 random-subset test. A song-selective electrode fires more strongly to song on average, but the seven song contacts as a set do not sit in the extreme tail of size-matched non-song draws. The song electrodes do show two secondary patterns: their representation looks somewhat more temporally stable (Fig. 3, panel b), and mean leave-one-out contributions to time-mean decoding are slightly larger for song-labeled contacts than for others ( $p = 0.014$ ), though effect sizes are small. Neither overturns the group-level null. The specialisation reported in Norman-Haignere et al. [10] is real and consistent with our data; it is just narrower than a naive reading would suggest.

**Acoustic vs. post-acoustic coding.** Subtracting out low-level acoustic information from the neural features removes about two-thirds of the song-vs-music RDM peak but leaves a residual (fraction surviving 0.33–0.47). Most of what the brain is doing in this distinction, then, really is just tracking basic sound structure. The permutation test on the residual does not cleanly clear significance at this  $N$ :  $p = 0.052$  for `song_only`,  $p = 0.108$  for `all`,  $p = 0.410$  for `no_song`. We therefore cannot claim a robustly significant non-acoustic component, and we should not call the surviving structure a “post-acoustic code.” What we can say is that some geometry remains after regressing out the specific linear acoustic feature set of Norman-Haignere et al. [10], that the surviving fraction is similar across subsets, and that the partialled peak arrives later in the trial (1.37–1.47 s, compared to raw peaks at 0.37–1.72 s). These are the directions expected if the STG combines basic acoustic input with higher-order category cues [2, 6–8, 11, 12], but they are suggestive rather than conclusive.

**Does it hold in continuous music?** The Bellier extension plays two roles. First, it is a naturalistic sanity check [4]: a vocal-vs-instrumental decoder reaches 0.78 balanced accuracy on 191 s of continuous song, and right STG slightly beats left STG in the same direction as Bellier et al. [1] (we did not run a permutation null on the anatomical comparison, so this is directional only). Second, the cross-dataset temporal-profile comparison shows that the two datasets emphasise different phases of the response. Norman’s trial-locked profiles highlight the sustained phase, while Bellier’s vocal-onset profiles highlight the transient onset. The “song-tracking” effect visible in Norman-Haignere et al. [10]’s 2-second clips is therefore best understood as a sustained post-onset response rather than a short onset transient. The two paradigms sample different halves of the same underlying response, which is expected given how differently the stimuli are presented.

**Caveats.** (i) *Sample size.* 33 Norman electrodes and 49 stimuli is small. Our partialled-divergence  $p$ -values sit near 0.05 because  $N$  is modest, not because the effect is obviously tiny. The signed direction and subset ordering are stable across bootstraps and shuffles; the absolute significance is under-powered. (ii) *Acoustic feature bank.* We partial out the exact regressors Norman-Haignere et al. [10] use. Richer audio embeddings (e.g. HuBERT, self-supervised music encoders) would absorb more

of the raw signal, so our surviving fraction is a conservative upper bound on the non-acoustic residual. (iii) *No permutation null on several Bellier comparisons.* We do not have a permutation null on the Bellier balanced-accuracy numbers or on the cross-temporal-generalisation stability gap in §3.6. The right-vs-left-STG ordering and the “more stable code on song electrodes” observation are therefore directional rather than formally significant. Adding these nulls is the single most fixable gap in the paper and is listed in Future Directions. (iv) *Song-electrode labels are author-supplied and defined on the same dataset.* We inherit the component assignments from Norman-Haignere et al. [10] rather than re-deriving them on held-out stimuli. The matched random-subset control does not require the labels to be “correct,” and the null result cannot be driven by the labels being informative – if they were, the song electrodes would sit in the upper tail of the random-draw distribution rather than near its middle. But a noisy or mis-specified label set could also produce the same null. A stronger follow-up would be to re-derive song-selectivity on a held-out stimulus split and then apply the random-subset control on the out-of-sample labels.

**Future directions.** Four natural next steps, in rough order of payoff for the effort required: (i) add label-shuffle permutation nulls to the Bellier decoder (for both the overall bacc and the right-vs-left-STG subset comparison) and to the cross-temporal generalisation gap; this is the cheapest upgrade that would turn several currently directional claims into formally significant ones; (ii) re-derive song-selectivity on a held-out stimulus split and re-apply the random-subset control on the out-of-sample labels, which would close the residual circularity in Caveat (iv); (iii) replace the acoustic regressor bank with a modern self-supervised audio embedding to push the surviving-fraction estimate in §3.5 closer to its true upper bound; (iv) apply the partialled-divergence pipeline to the Bellier stimulus windows to ask whether the sustained vocal-tracking component that emerges in the continuous song also survives acoustic partialling.

**Novelty.** Our contributions on top of the published work are the matched random-subset control, the acoustic-partialled divergence pipeline, and the cross-dataset temporal-profile comparison (together with the interleaved-blocked-folds CV fix on Bellier). Taken together, these test whether the “song-selective population exists” result also implies disproportionate information content, and find that it does not.

**Code and data availability.** All analyses run from the `logic/` directory (`python pipeline.py`, random seed 42) or via `python scripts/generate_composite_figures.py` from the repository root for the multi-panel figures (see `--help`, including `--only 3` for the Bellier composite alone). The full codebase, including the notebook used to generate every figure and table, is available at <https://github.com/djordjeivanovic11/neuro120-project>. A separate document states course policy on generative-AI use. Norman-Haignere data are from the public release accompanying Norman-Haignere et al. [10]; Bellier ECoG data are from the public release accompanying Bellier et al. [1].

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## Appendix A: Project Breakdown

- **Djordje Ivanovic.** Code and modeling
- **Georgia Hutchinson.** modeling and presentation of results, methods
- **JaKayla Harris.** Introductory research, research survey

All three members jointly scoped the research questions, agreed on the matched random-subset control and the acoustic-partialled extension, and reviewed every figure and table in this manuscript.